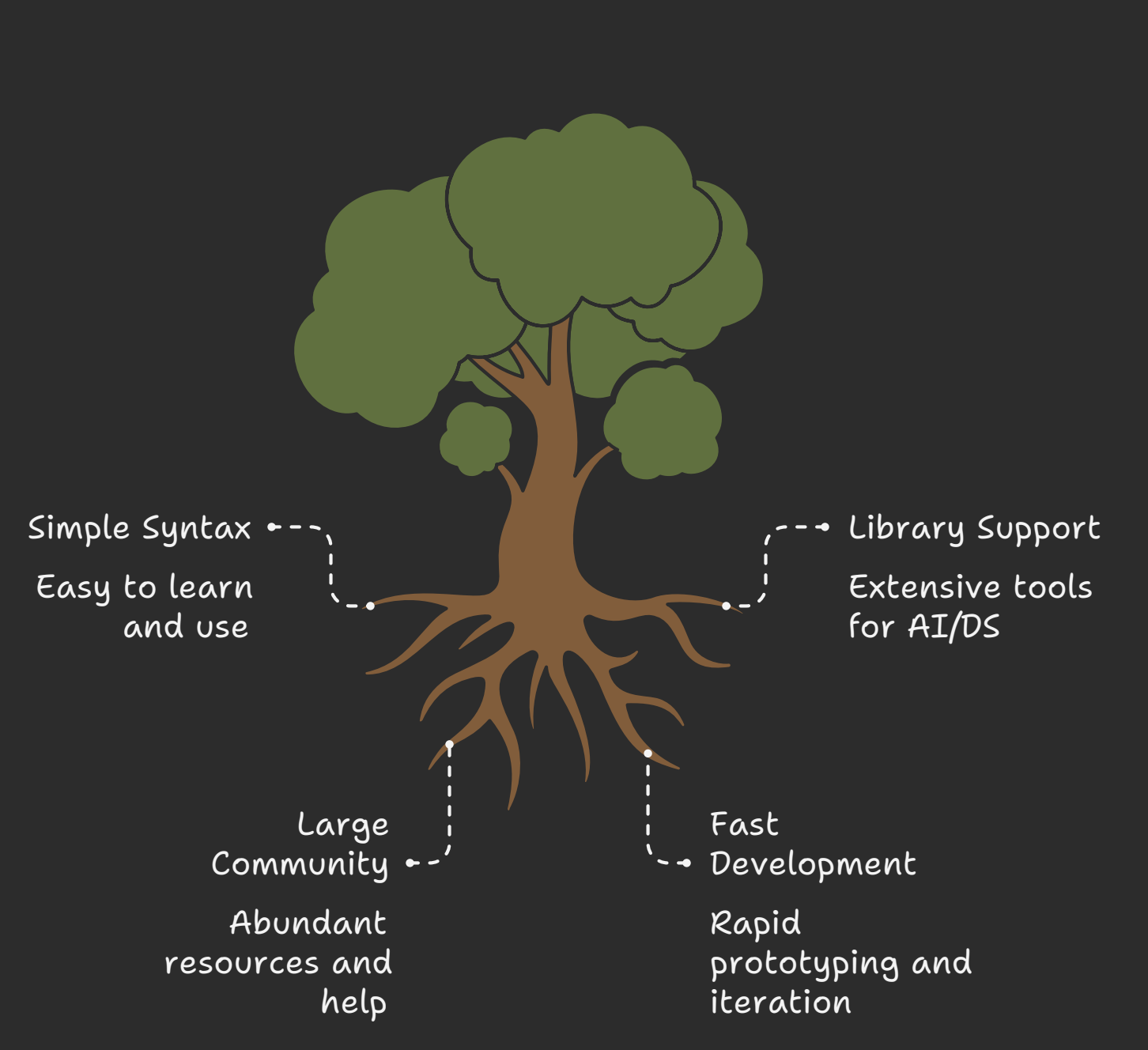
Day 1: Machine Learning

PART 1: Python Basics

1] What is Python? Why is it popular in AI & Data Science?

Python's Popularity in AI & Data Science


Simple Syntax
Easy to learn and use
Library Support
Extensive tools for AI/DS
Large Community
Abundant resources and help
Fast Development
Rapid prototyping and iteration

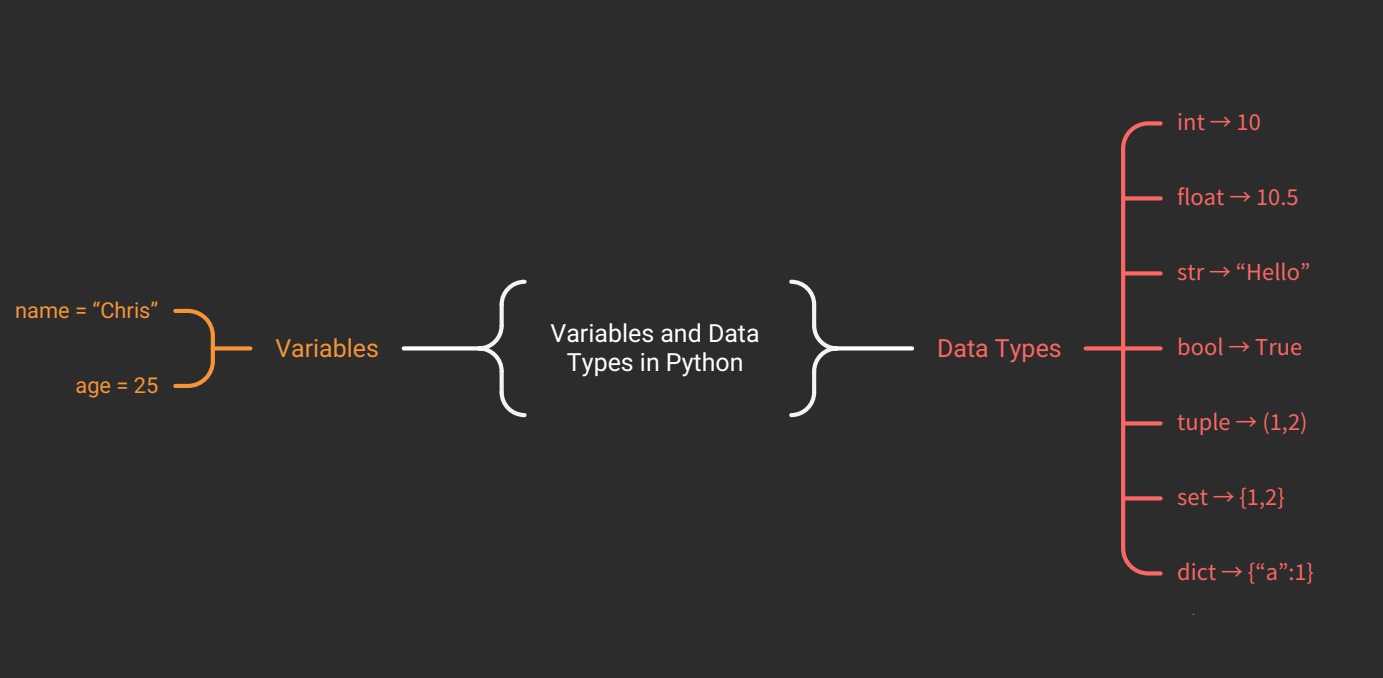
2] Difference between List, Tuple, Set and Dictionary

Comparison of Python Data Structures

Characteristic	List	Tuple	Set	Dictionary
Order	Ordered	Ordered	Unordered	Key-value pairs
Mutability	Mutable	Immutable	Mutable	Mutable
Duplicates	Allows duplicates	Allows duplicates	No duplicates	Keys must be unique
Syntax	[ ]	( )	{ }	{key:value}
Example	nums = [1, 2]	nums = (1, 2)	colors = {“red”, “blue”}	student = {“name”: “John”}

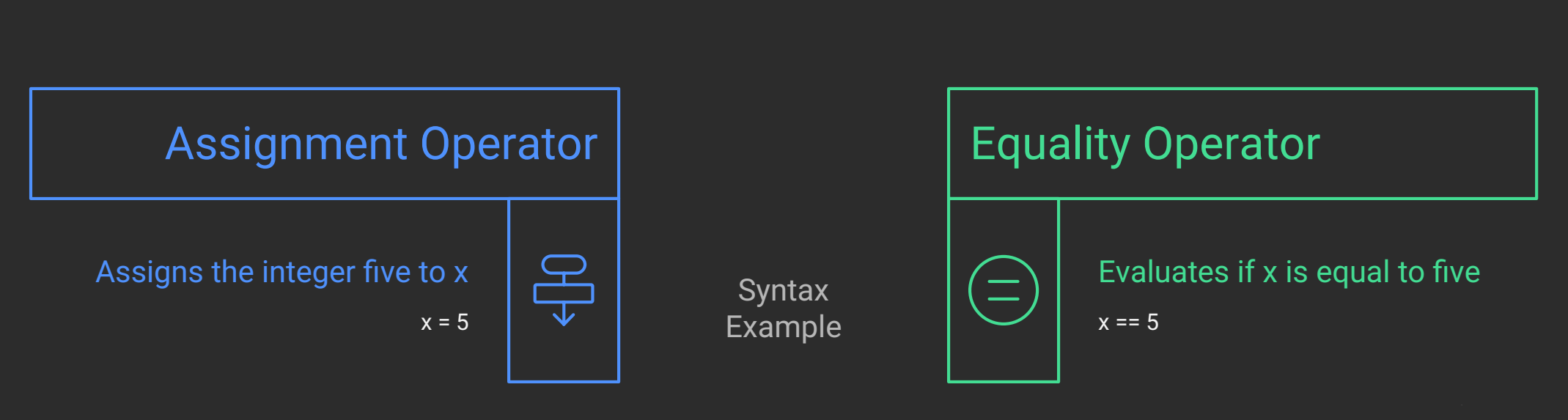
3] Variables and Data Types in Python

Variables and Data Types in Python


name = “Chris”
age = 25
Variables
Variables and Data Types in Python
Data Types
int → 10
float → 10.5
str → “Hello”
bool → True
tuple → (1,2)
set → {1,2}
dict → {“a”:1}

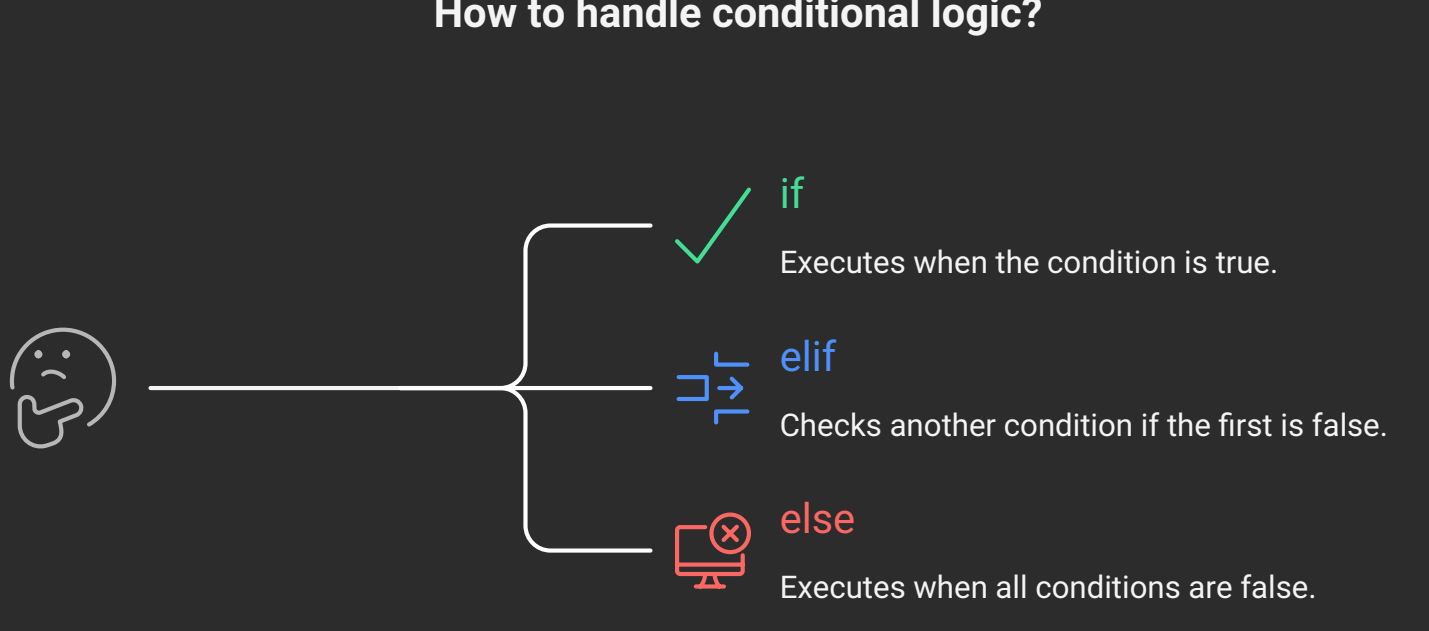
4] Difference between = and ==

Which operator should I use in my code?


Assignment Operator
Assigns the integer five to x
x = 5
Syntax Example
Equality Operator
Evaluates if x is equal to five
x == 5

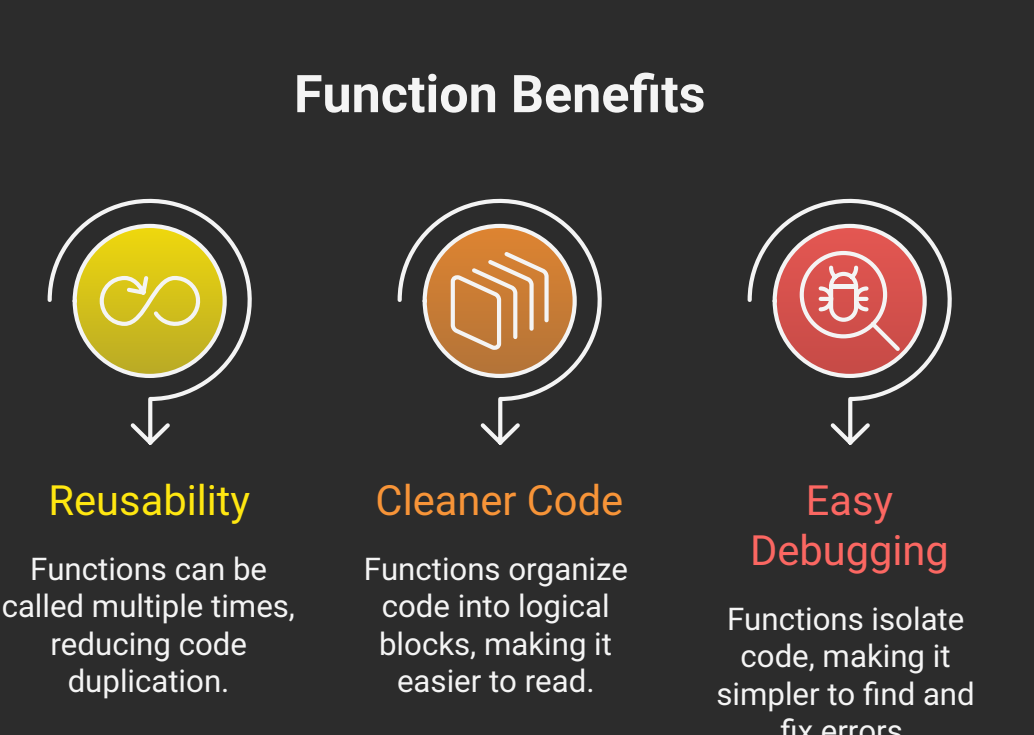
5] if, elif and else

How to handle conditional logic?


if
Executes when the condition is true.
elif
Checks another condition if the first is false.
else
Executes when all conditions are false.

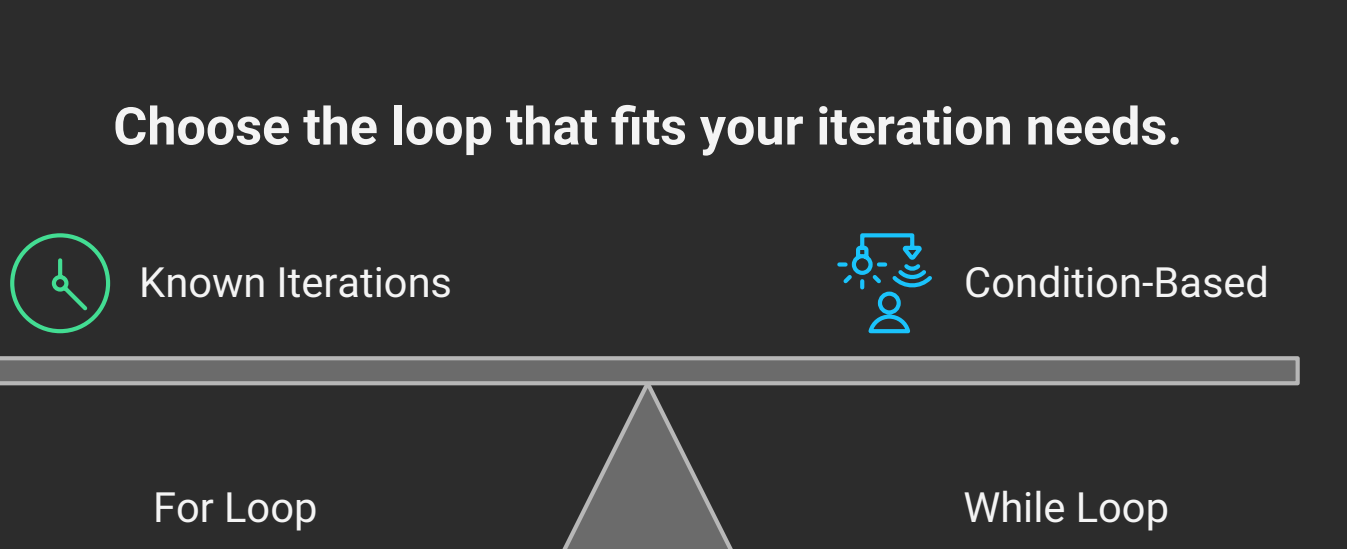
6] What is a Function?

Function Benefits


Reusability
Functions can be called multiple times, reducing code duplication.
Cleaner Code
Functions organize code into logical blocks, making it easier to read.
Easy Debugging
Functions isolate code, making it simpler to find and fix errors.

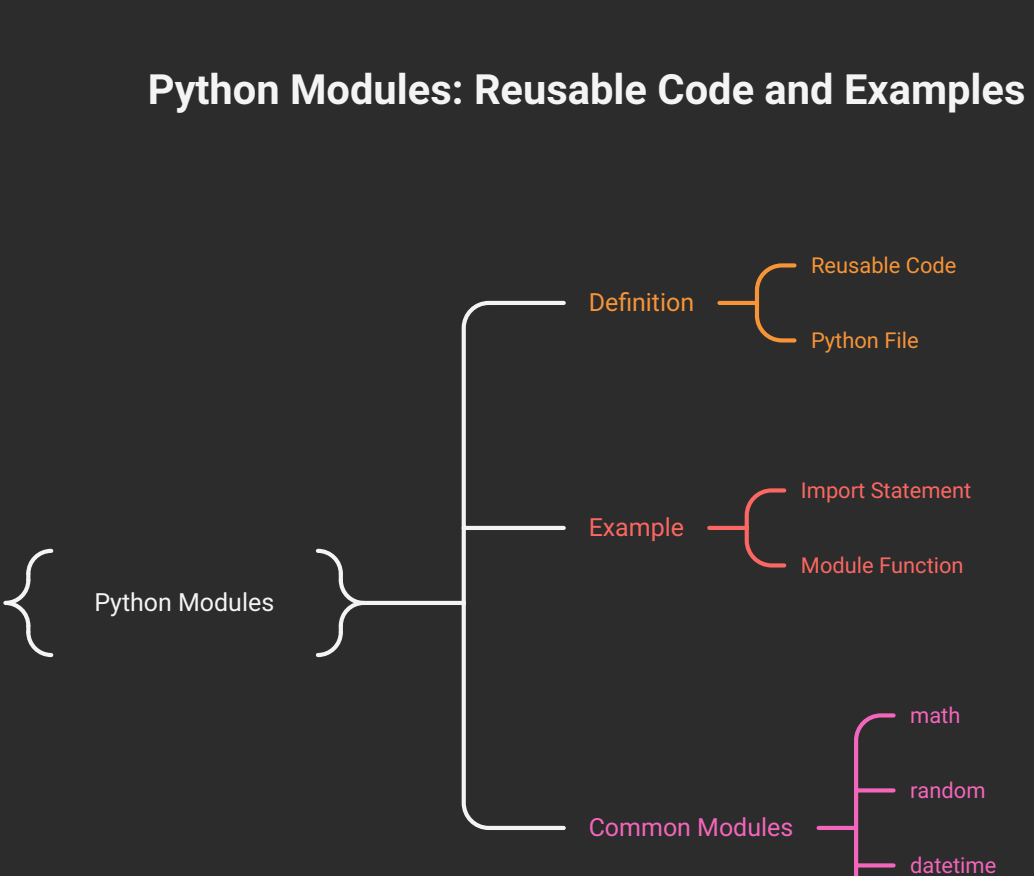
7] Difference between for loop and while loop

Choose the loop that fits your iteration needs.


Known Iterations
Condition-Based
For Loop
While Loop

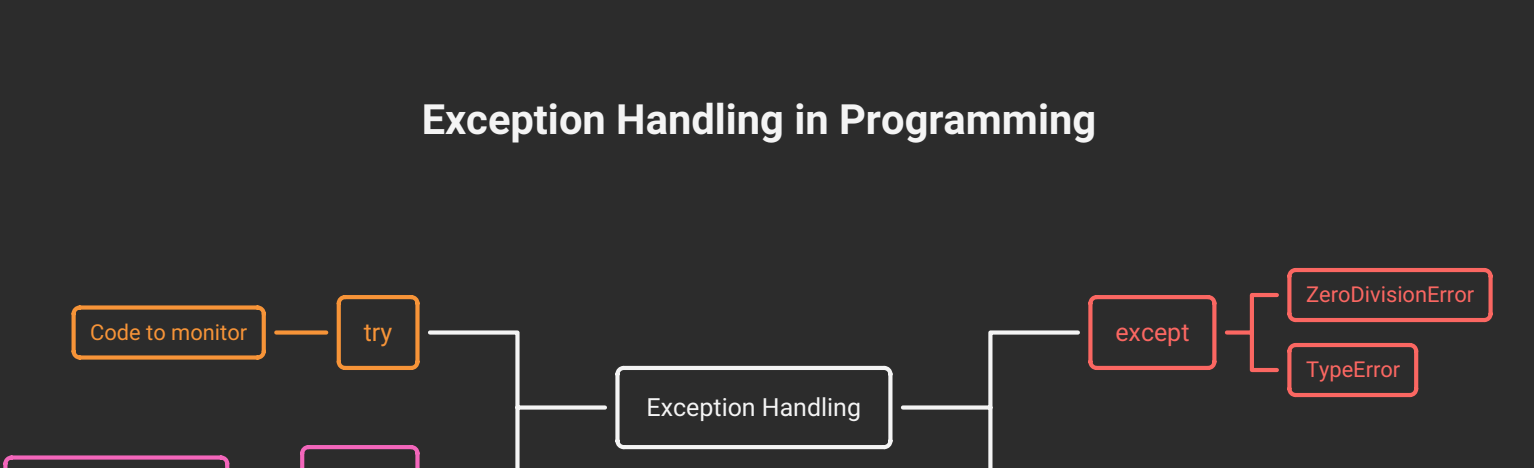
8] What is a Module?

Python Modules: Reusable Code and Examples


Python Modules
Definition
Reusable Code
Python File
Example
Import Statement
Module Function
Common Modules
math
random
datetime
os

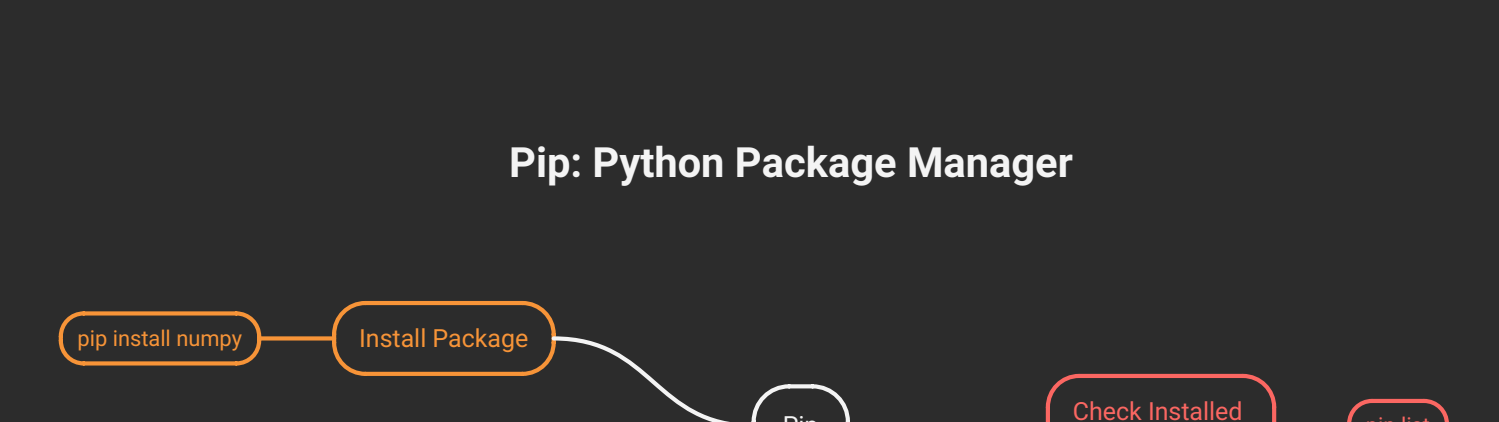
9] What is Exception Handling?

Exception Handling in Programming


Code to monitor
try
Cleanup operations
finally
Exception Handling
except
ZeroDivisionError
TypeError
raise
Custom exceptions

10] What is pip?

Pip: Python Package Manager

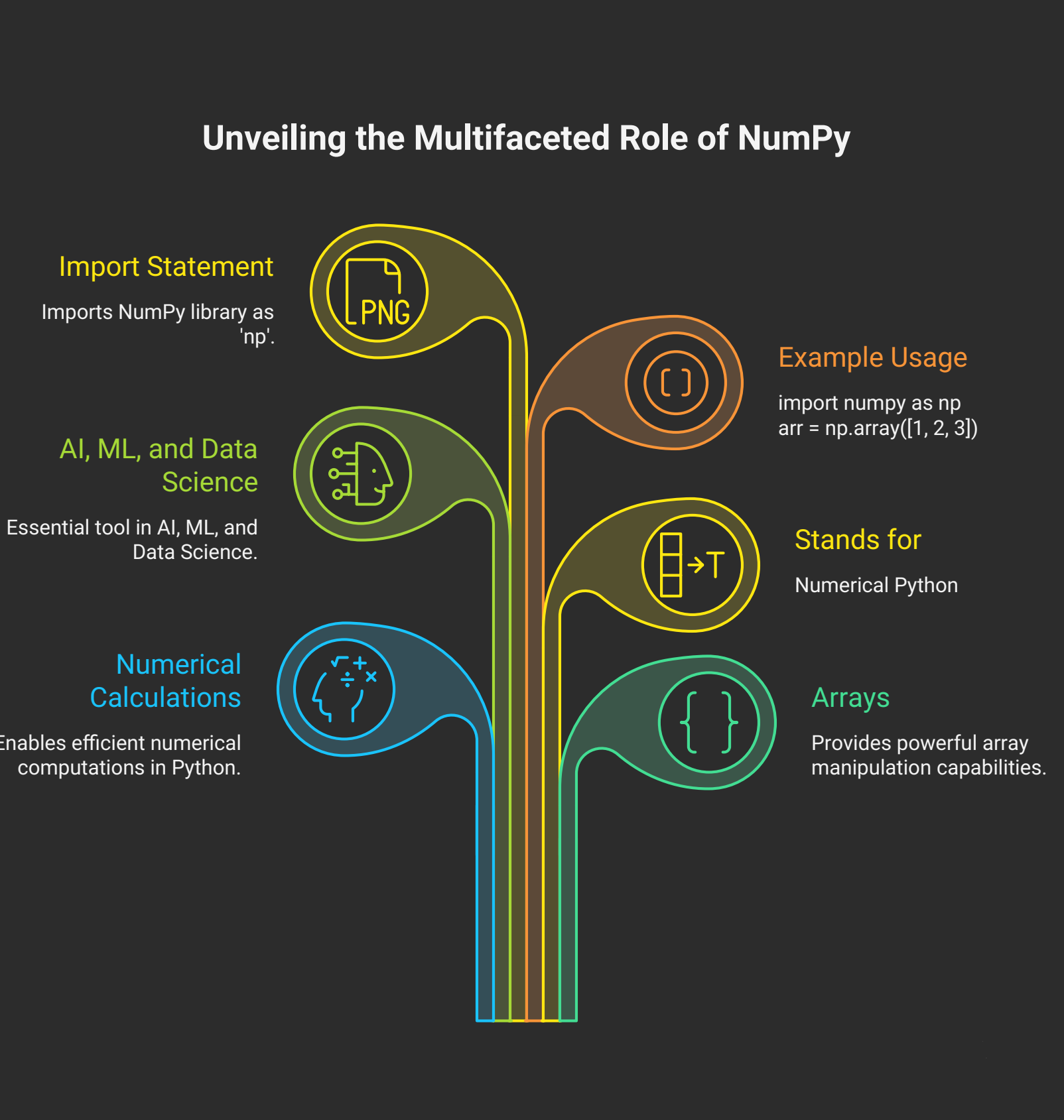

pip install numpy
Install Package
pip install –upgrade numpy
Upgrade Package
Pip
Check Installed Packages
pip list

PART 2: NumPy Basics

1] What is NumPy?

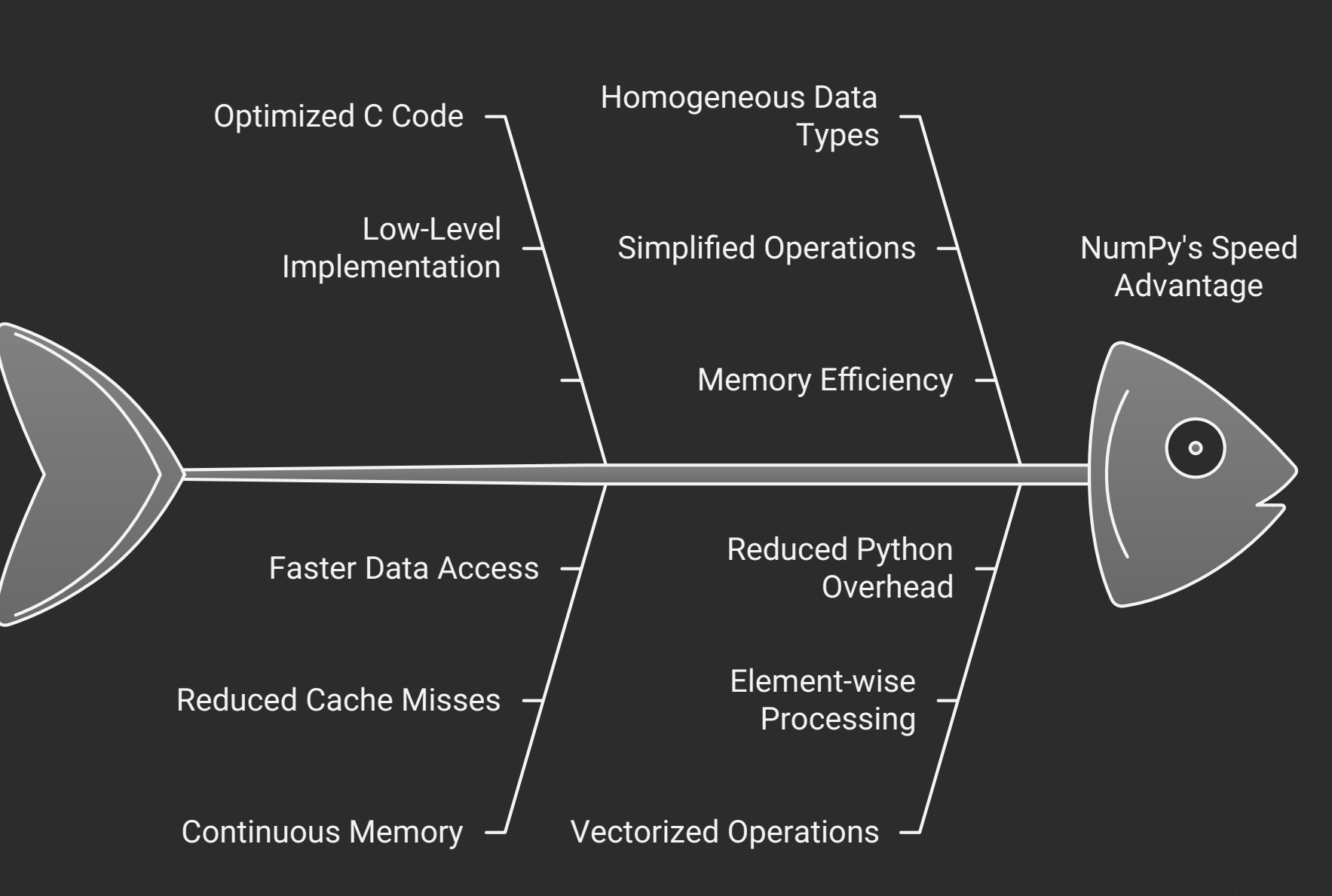
- NumPy stands for Numerical Python.
- It is a Python library used for numerical calculations and arrays.
- Widely used in AI, Machine Learning and Data Science.

Unveiling the Multifaceted Role of NumPy


Import Statement
Imports NumPy library as 'np'.
AI, ML, and Data Science
Essential tool in AI, ML, and Data Science.
Numerical Calculations
Enables efficient numerical computations in Python.
Example Usage
import numpy as np
arr = np.array([1, 2, 3])
Stands for
Numerical Python
Arrays
Provides powerful array manipulation capabilities.

2. Why is NumPy faster than Python lists?

Why NumPy is Faster


Optimized C Code
Low-Level Implementation
Homogeneous Data Types
Simplified Operations
Memory Efficiency
NumPy's Speed Advantage
Faster Data Access
Reduced Cache Misses
Continuous Memory
Reduced Python Overhead
Element-wise Processing
Vectorized Operations

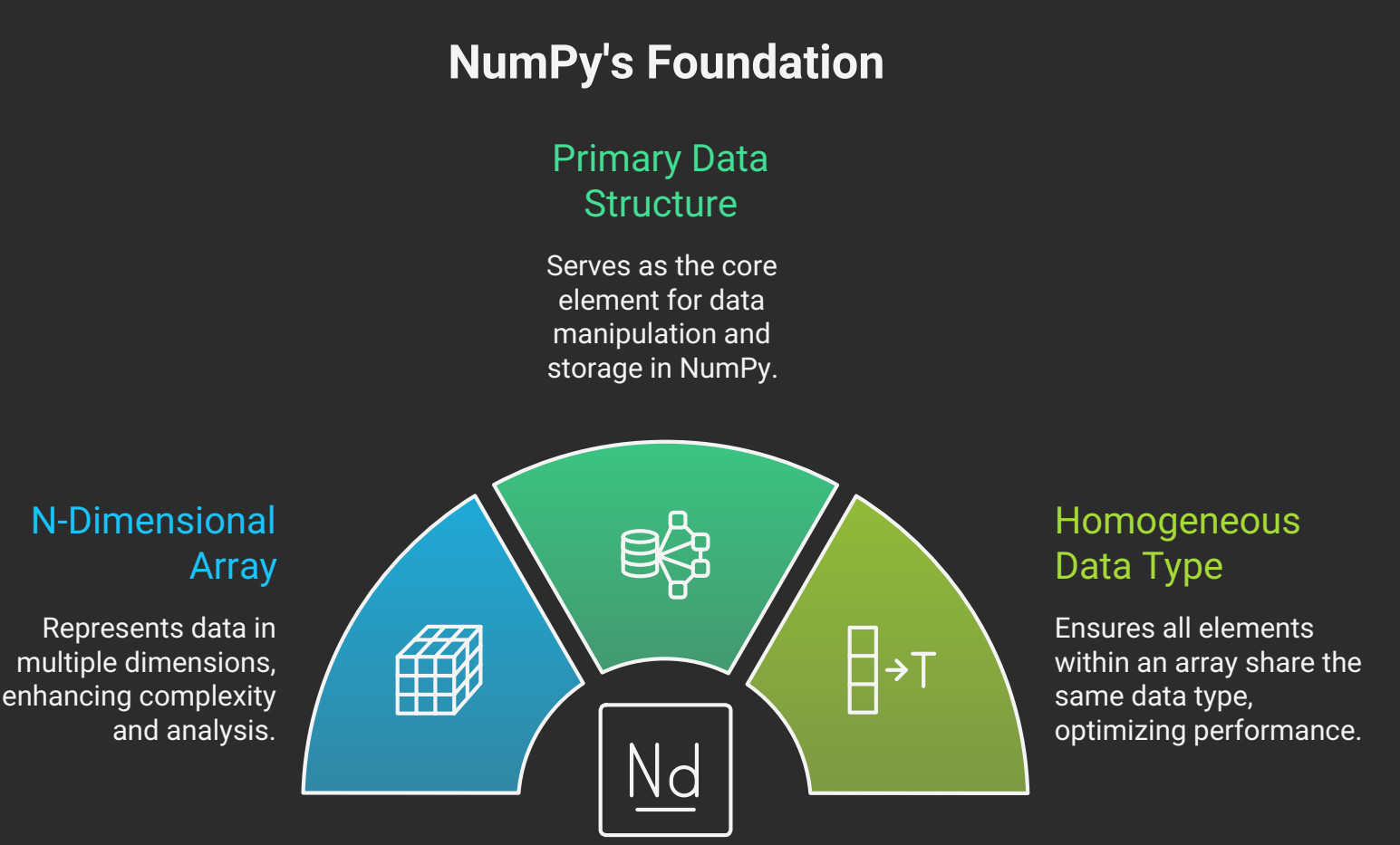
Example:
a = np.array([1, 2, 3]) b = np.array([4, 5, 6])

print(a + b)

Output:[5 7 9]

3] What is an ndarray?

NumPy's Foundation

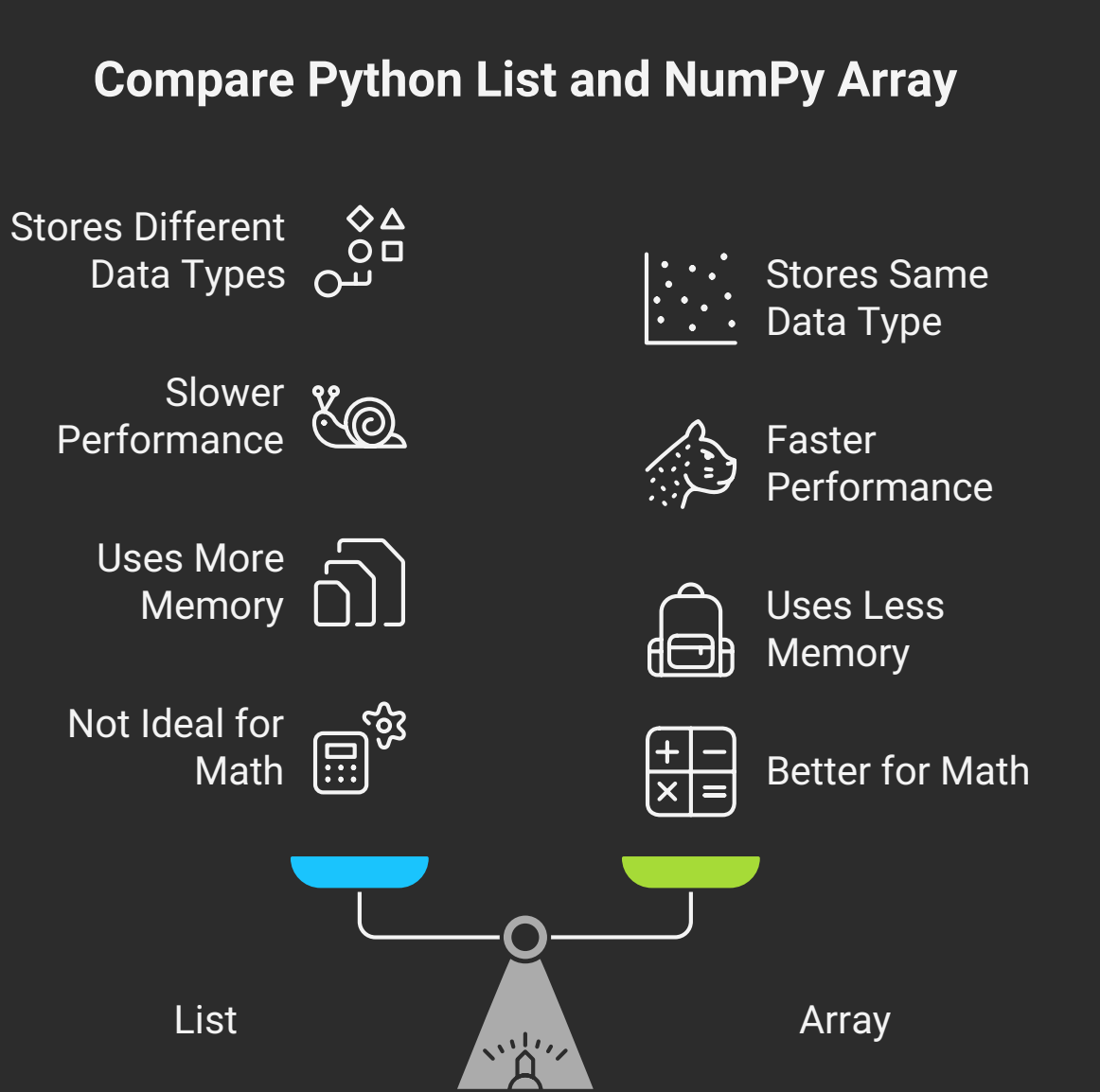

Primary Data Structure
Serves as the core element for data manipulation and storage in NumPy.
N-Dimensional Array
Represents data in multiple dimensions, enhancing complexity and analysis.
Homogeneous Data Type
Ensures all elements within an array share the same data type, optimizing performance.

Example:arr = np.array([1, 2, 3])

2D Array Example:arr = np.array([[1, 2], [3, 4]])

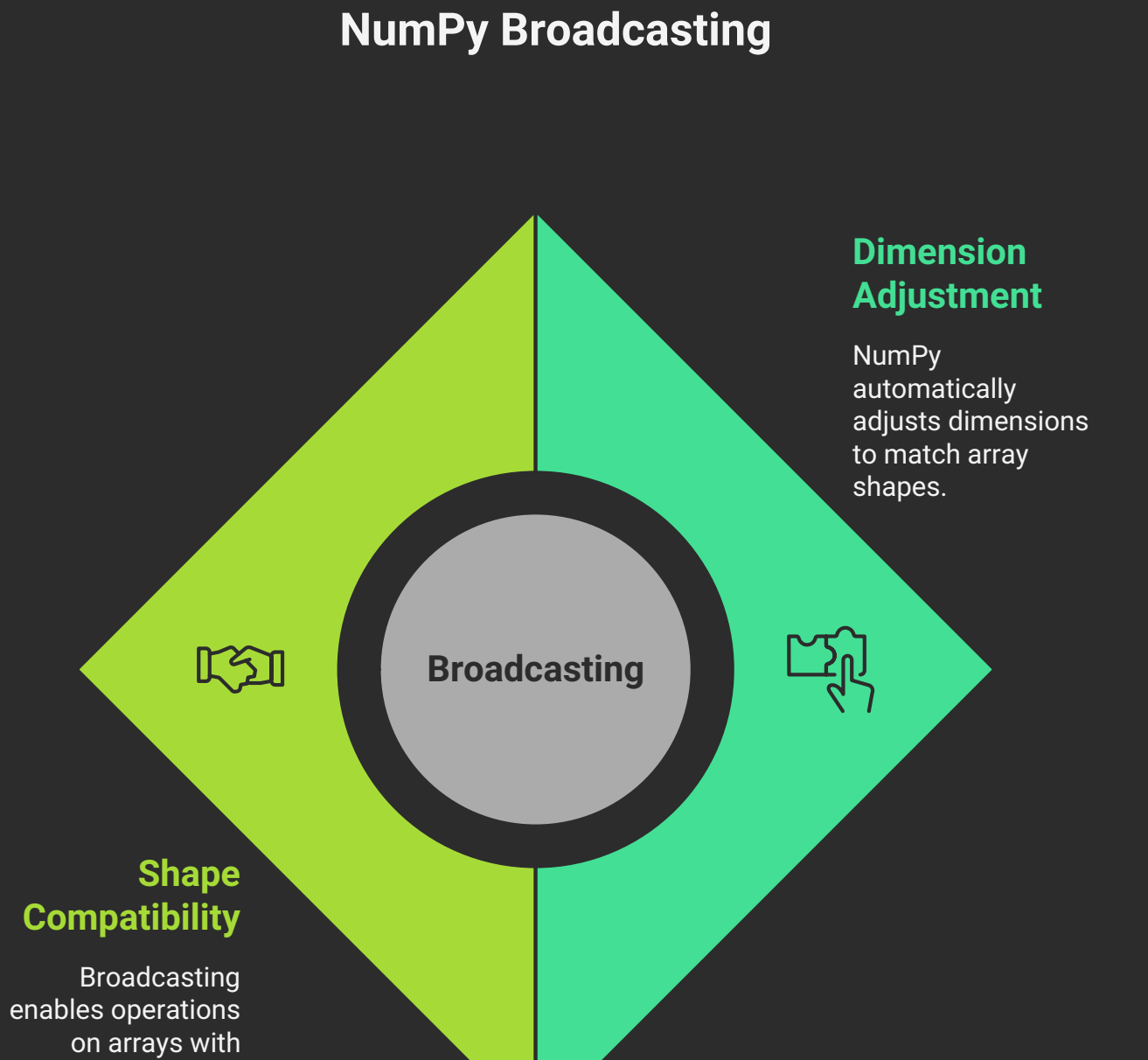
4] Difference between Array and List

Compare Python List and NumPy Array


Stores Different Data Types
Slower Performance
Uses More Memory
Not Ideal for Math
Stores Same Data Type
Faster Performance
Uses Less Memory
Better for Math
List
Array

5] What is Broadcasting?

NumPy Broadcasting


Dimension Adjustment
NumPy automatically adjusts dimensions to match array shapes.
Broadcasting
Shape Compatibility
Broadcasting enables operations on arrays with different shapes.

Example:arr = np.array([1, 2, 3])

print(arr + 5)
Output:[6 7 8]

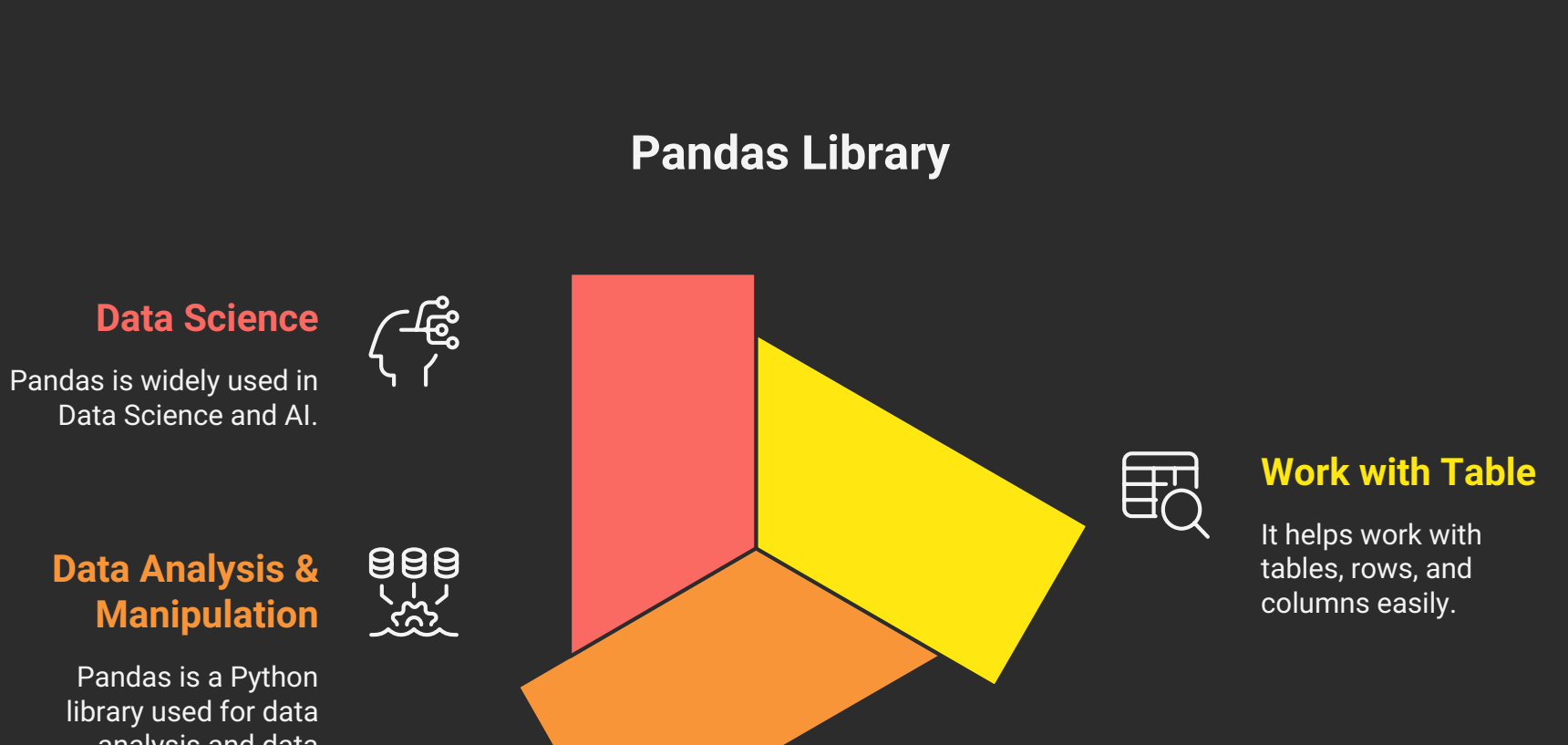
Here, 5 is added to every element of the array.

PART 3: Pandas Basics

1] What is Pandas?

- Pandas is a Python library used for data analysis and data manipulation.
- It helps work with tables, rows and columns easily.
- Widely used in Data Science and AI.

Pandas Library


Data Science
Pandas is widely used in Data Science and AI.
Data Analysis & Manipulation
Pandas is a Python library used for data analysis and data manipulation.
Work with Table
It helps work with tables, rows, and columns easily.

Example:import pandas as pddata = pd.DataFrame()

2] What is a DataFrame?

DataFrame Characteristics

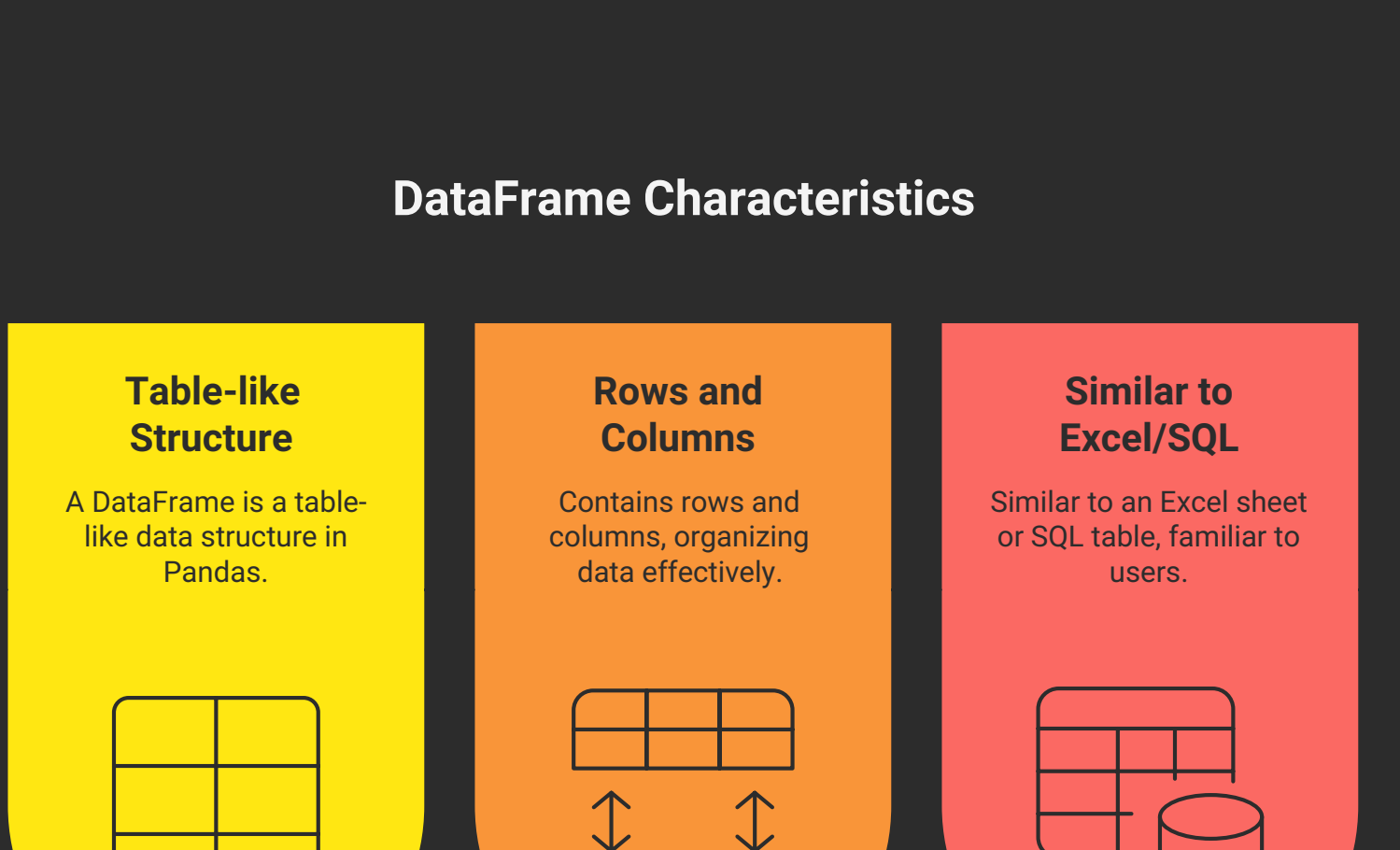


Table-like Structure
A DataFrame is a table-like data structure in Pandas.
Rows and Columns
Contains rows and columns, organizing data effectively.
Similar to Excel/SQL
Similar to an Excel sheet or SQL table, familiar to users.

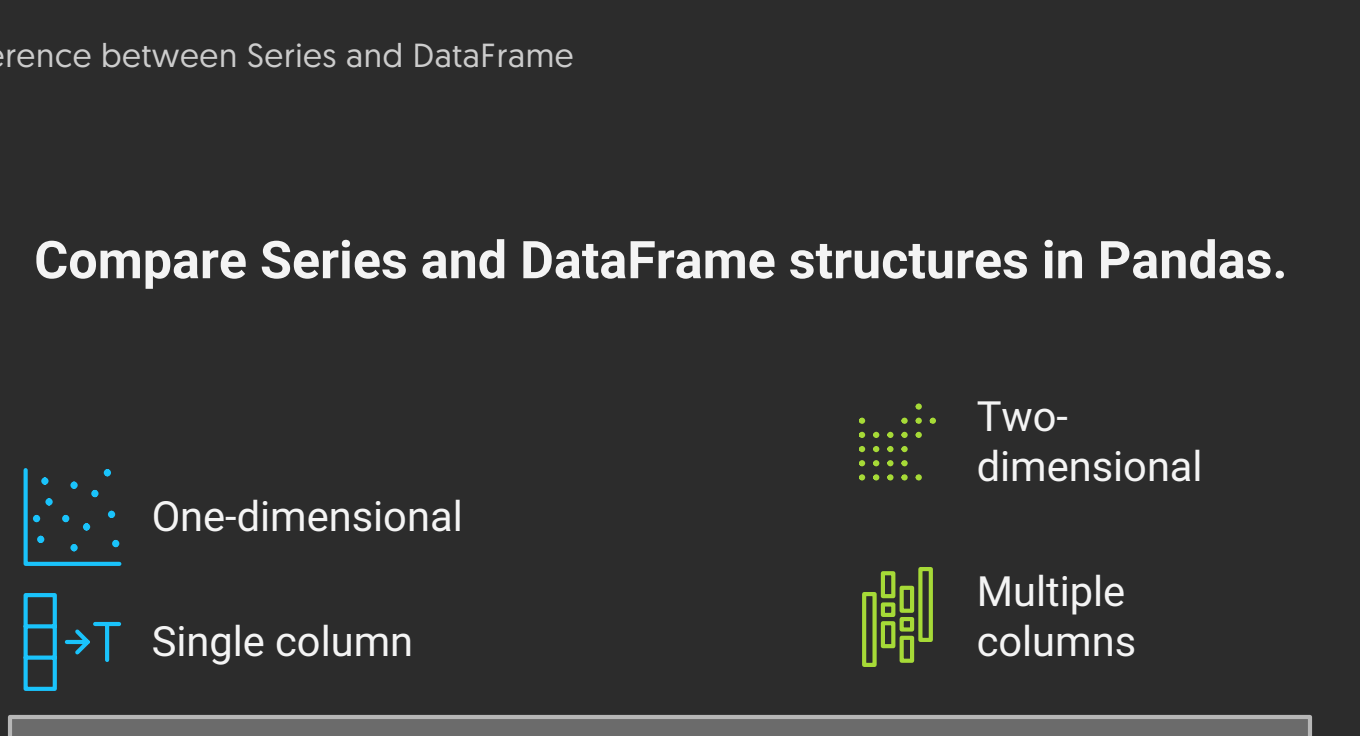
Example:data = {   “Name”: [“John”, “Alex”],   “Age”: [20, 22]   }

df = pd.DataFrame(data)

Example Table:
Name   AgeJohn   20Alex   22

3] Difference between Series and DataFrame

Compare Series and DataFrame structures in Pandas.

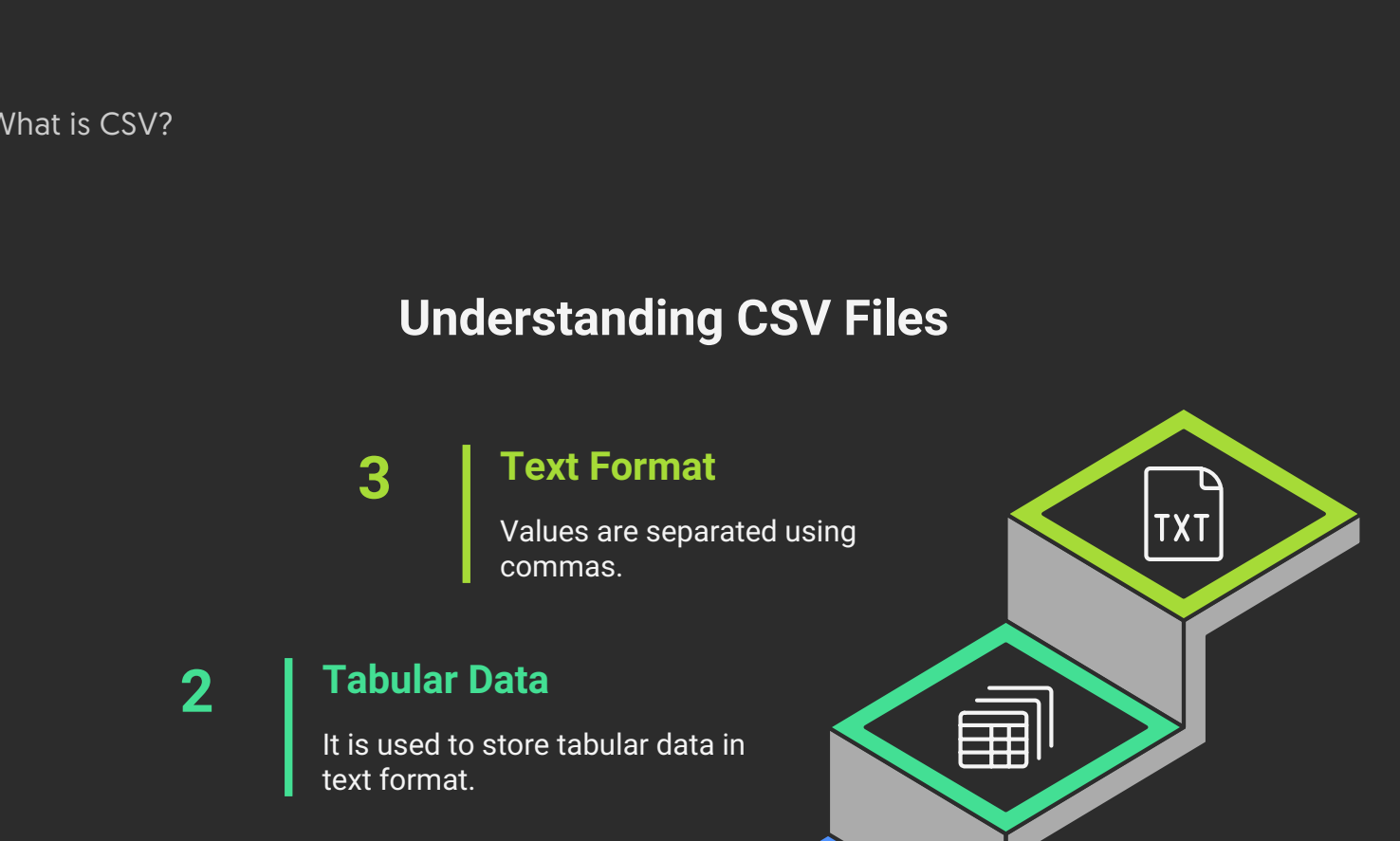

One-dimensional
Single column
Two-dimensional
Multiple columns
Series
DataFrame

Series Example:s = pd.Series([10, 20, 30])

DataFrame Example:df = pd.DataFrame({   “A”: [1, 2],   “B”: [3, 4]   })

4. What is CSV?

Understanding CSV Files


3
Text Format
Values are separated using commas.
2
Tabular Data
It is used to store tabular data in text format.
1
Comma Separated Values
CSV stands for Comma Separated Values.

Example CSV:Name,AgeJohn,20Alex,22

Read CSV using Pandas:df = pd.read_csv(“data.csv”)

5] Why Pandas is important for AI?

Pandas is important because it helps:

- Clean data
- Organise data
- Analyse data
- Handle missing values
- Prepare datasets for Machine Learning

Common tasks:

- Reading CSV files
- Filtering data
- Sorting data
- Data preprocessing

Example:df.head()

Used to view first 5 rows of data.